

Lesson 2: Data Preprocessing and Algorithm Selection



2.0 Lesson Overview



Lesson Overview

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2.1 Exploring the Data and Replacing Incorrect Values



Exploring the Data



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Exploring the Data

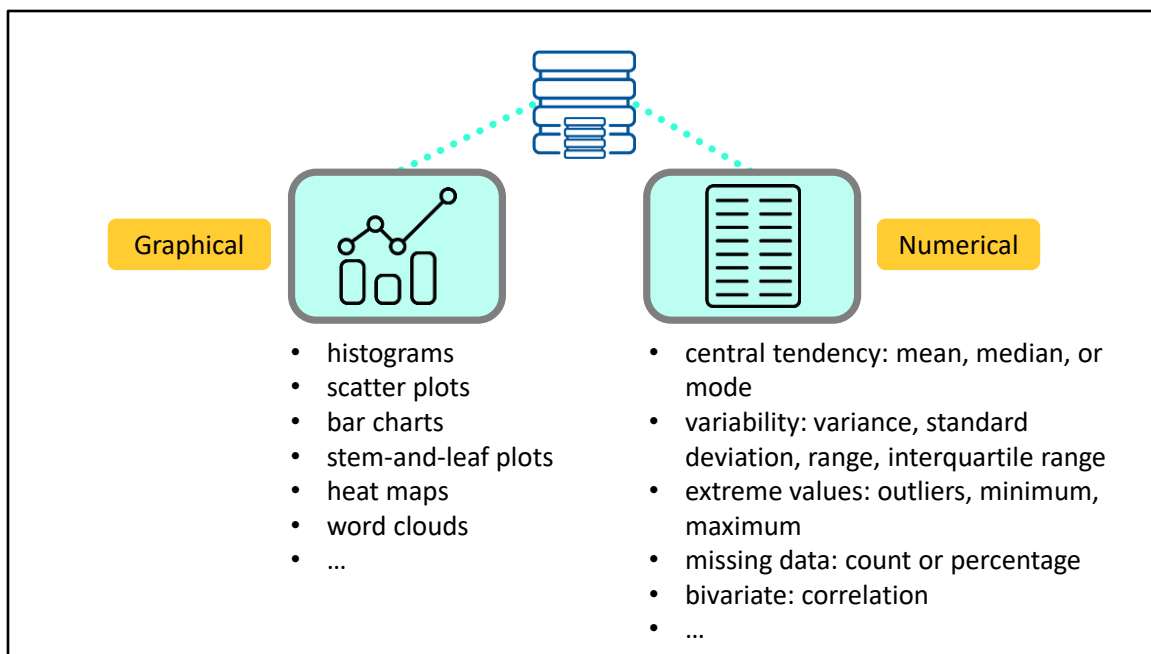


Essential Data Tasks

- ✓ Divide the data.
- ✓ Address rare events.
- ✓ Manage missing values.
 - Replace incorrect values.
 - Add unstructured data.
 - Extract features.
 - Handle extreme or unusual values.
 - Select useful inputs.



In any data analysis, it's important to get to know your data. Exploring the data is one of the most important steps of an analytical project, and it can be time-consuming. However, this is time well-spent. By gaining an intimate understanding of the variables and how they relate to one another, you can make better decisions about how to prepare the data.



You can explore the data using both graphical and numerical methods.

Common graphical tools include histograms, scatter plots, bar charts, and stem-and-leaf plots. These have been popular for decades, because they are easy to create by hand or with a computer, and they're easy to understand. More modern graphical tools, such as heat maps and word clouds, scale well to large data sets and help you make sense of bigger, more complex data.

Numerical methods used for data exploration include summary statistics for measures of central tendency, such as the mean, median, or mode; and statistics that measure variability, such as variance, standard deviation, range, or interquartile-range. You can investigate extreme values such as outliers, the minimum, or the maximum, as well as counts or percentages of missing data. And you can use bivariate measures such as correlation to estimate the association between two variables.

Demo: Exploring the Data

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Question: Data Exploration Node



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Question: Data Exploration Node

In Model Studio, the Data Exploration node is in which group?

- a. Data Preprocessing
- b. Miscellaneous
- c. Postprocessing
- d. Supervised Learning

Answer: Data Exploration Node

In Model Studio, the Data Exploration node is in which group?

- a. Data Preprocessing
- ☒ b. Miscellaneous
- c. Postprocessing
- d. Supervised Learning

CA: b

FB: The Data Exploration node is in the Miscellaneous group.

Data Preprocessing Nodes in Model Studio

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Risk Modeling Add-on for SAS Viya

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Replacing Incorrect Values



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Replacing Incorrect Values

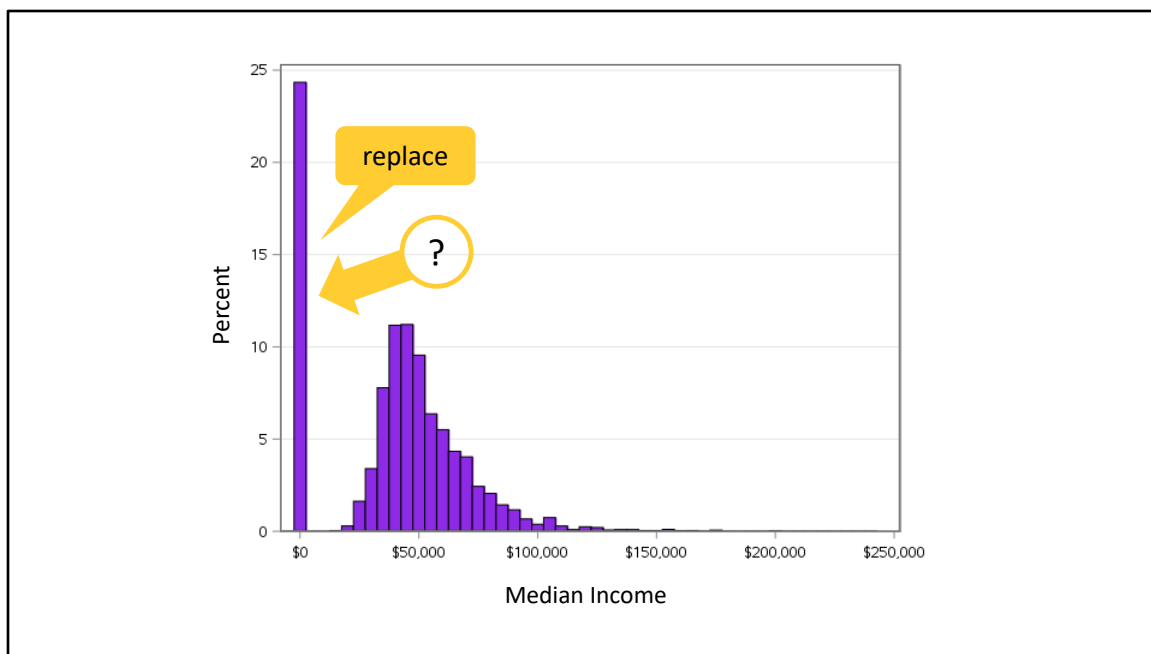


Essential Data Tasks

- ✓ Divide the data.
- ✓ Address rare events.
- ✓ Manage missing values.
 - Replace incorrect values.
 - Add unstructured data.
 - Extract features.
 - Handle extreme or unusual values.
 - Select useful inputs.



During data exploration, it is common to find that some data values are incorrect. Perhaps they don't make sense, or they are unrealistic. There are many ways to address this issue. The one we show you is to replace the incorrect values with more appropriate values.



Here is a histogram of income. Notice that there is a large spike of income values at 0. Could that many households in the data set actually have a value of 0 for income? Most likely, these values represent people who did not want to report their income. Instead of leaving a blank income field on a survey, those people might have written the value 0. Or, the data entry policy might have been to use a value of 0 for blank numeric fields.

In this situation, it makes sense to replace the 0 income values with more realistic numbers. Determining the best replacement values relies on business knowledge and familiarity with the data.

Demo: Replacing Incorrect Values Starting on the Data Tab

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Replacing Incorrect Values Starting with the Manage Variables Node

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2.2 Extracting Features



Feature Extraction



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Feature Extraction



Essential Data Tasks

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- ✓ Address rare events.
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- Add unstructured data.
- Extract features.
- Handle extreme or unusual values.
- Select useful inputs.

feature
extraction

The next two essential data tasks (adding unstructured data and extracting features) are both methods of feature extraction. Feature extraction means creating new input variables based on inputs that are already in the data.

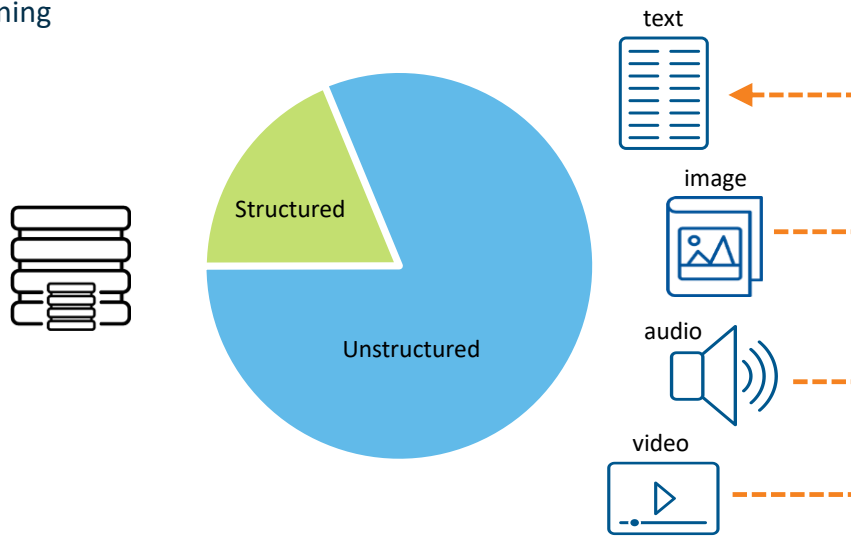
To add unstructured data, we convert unstructured text variables to usable numeric inputs. This involves text mining.

Text Mining



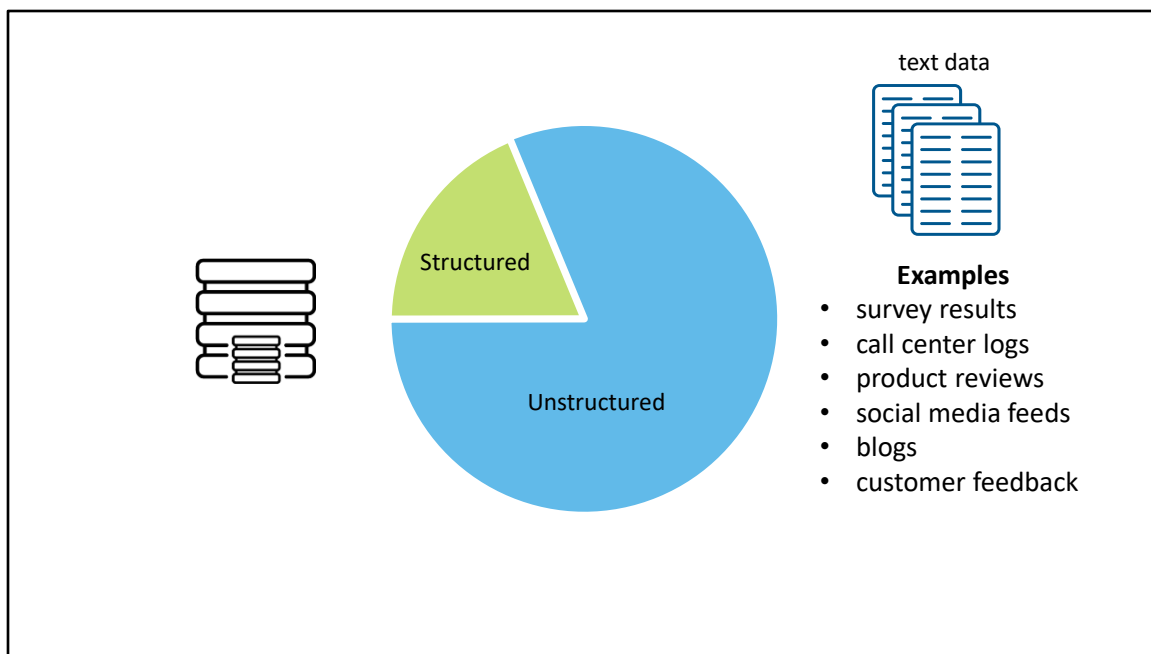
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Text Mining

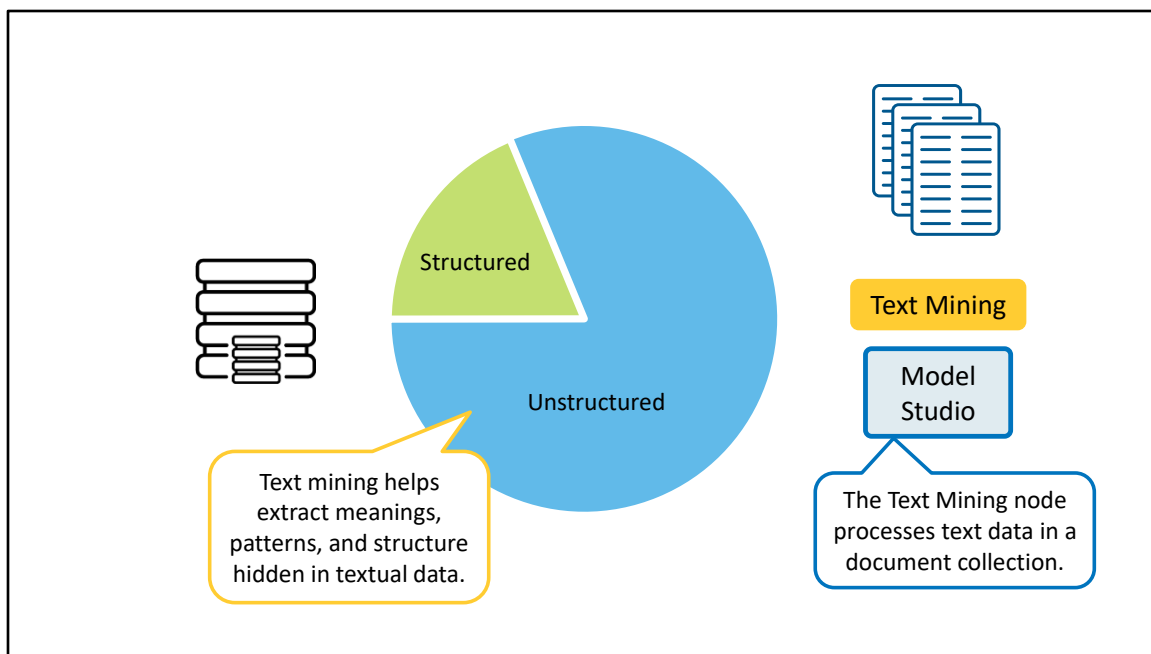


Organizations today are generating and storing tremendous amounts of data. A majority of that data is *unstructured* (that is, information that either does not have a predefined data model or is not organized in a predefined way).

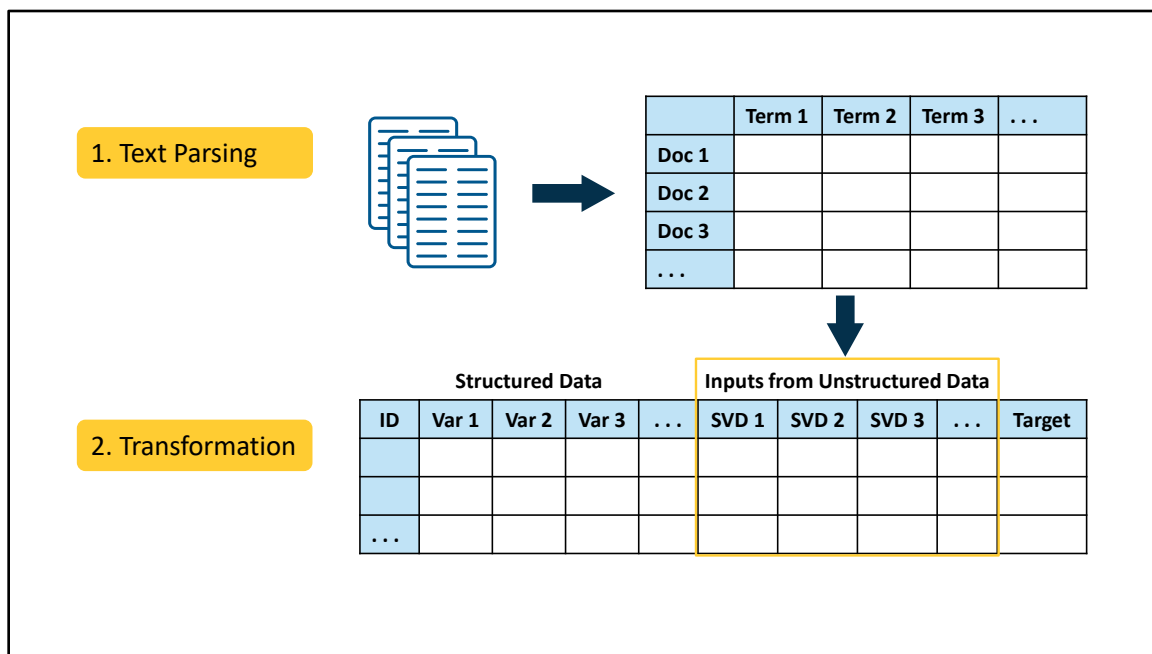
Unstructured data can be text or non-text (such as images, audio, or video). Non-textual data are often converted to text data, so we focus on text data in this course.



Unstructured text data are commonly available in business (for example, survey results, call center logs, product reviews, social media feeds, blogs, and customer feedback). These data contain information that is useful for predictive modeling outcomes, so it is important to analyze the unstructured text data sources along with the structured data.



Unlocking the full potential within these complex data sources can be tricky, but it is possible using text mining techniques. In Model Studio, the Text Mining node enables you to process text data in a document collection. Adding text mining results can improve the predictive ability of models that are based only on structured data.



In text mining, data are processed in two phases: text parsing and transformation. Text parsing processes textual data into a term-by-document frequency matrix. Transformations such as singular value decomposition (or SVD) change this matrix into a data set that is suitable for data mining purposes. As a result of text mining, a document collection with thousands of documents and terms can be represented in a compact and efficient form.

Singular Value Decomposition and the Text Mining Node

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Demo: Adding Text Mining Features

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Question: Text Mining Node



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Question: Text Mining Node

Which statement is true about the new input columns that the Text Mining node creates?

- a. The Text Mining node creates a new input column for each topic that it discovers.
- b. The Text Mining node creates a new input column for each term within each topic.
- c. The measurement scale of each new input column is ordinal.
- d. Each new input column's name starts with a plus sign (+)

Answer: Text Mining Node

Which statement is true about the new input columns that the Text Mining node creates?

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- b. The Text Mining node creates a new input column for each term within each topic.
- c. The measurement scale of each new input column is ordinal.
- d. Each new input column's name starts with a plus sign (+)

CA: a

FB: The Text Mining node creates a new input column for each topic it discovers.

Feature Extraction Node

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2.3 Transforming Inputs



Using Transformations to Handle Extreme or Unusual Values



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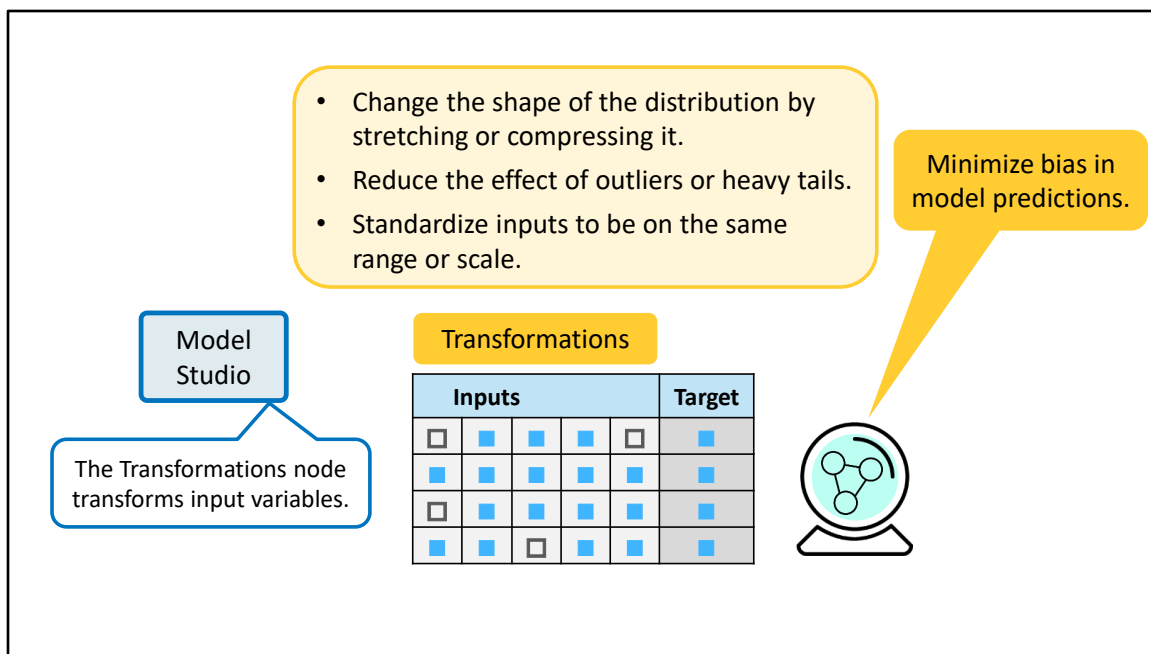
Using Transformations to Handle Extreme or Unusual Values



Essential Data Tasks

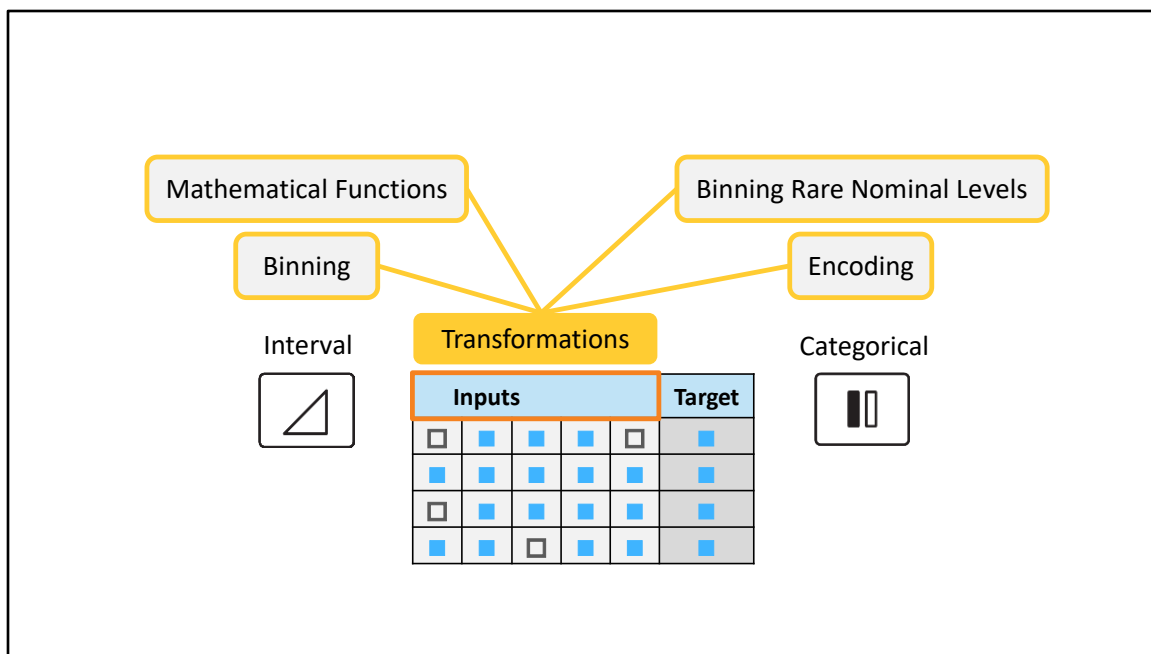
- ✓ Divide the data.
 - ✓ Address rare events.
 - ✓ Manage missing values.
 - ✓ Replace incorrect values.
 - ✓ Add unstructured data.
 - ✓ Extract features.
 - **Handle extreme or unusual values.**
 - Select useful inputs.
- transformations

Extreme or unusual values of the input variables can lead to problems in predictive modeling. These values might be outliers or they might occur in variables whose distributions are highly skewed (that is, not symmetric). Such values can lead to biased predictions from machine learning models, so it is a common practice to manage these values during data preprocessing. The solution is to apply transformations to the input variables.



Transformations can minimize bias in model predictions in various ways. Transformations are most commonly used to change the shape of the distribution of a variable by stretching or compressing it, reduce the effect of outliers or heavy tails, and standardize inputs to be on the same range or scale.

When you prepare data for modeling, you can use the Transformations node to transform input variables.



You choose from different transformation methods, depending on whether the input variable is interval or categorical. For interval inputs it is common to use transformations based on either mathematical functions or binning. For categorical inputs, you can bin rare nominal levels or apply various types of encoding.

Let's take a closer look at the transformations for interval inputs.

Mathematical Functions

- Centering
- Exponential
- Inverse
- Inverse Square
- Inverse Square Root
- $\text{Log}_e / \text{Log}_{10}$
- Range
- Square
- Square Root
- Standardize

Interval



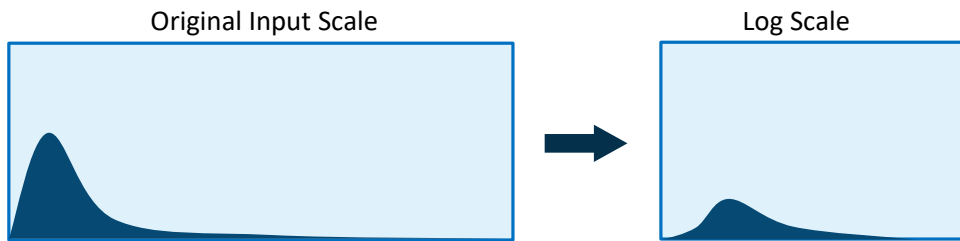
Transformations

Inputs					Target
☐	☒	☒	☒	☐	☒
☒	☒	☒	☒	☒	☒
☐	☒	☒	☒	☒	☒
☒	☒	☐	☒	☒	☒

Examples of mathematical functions are centering, exponential, inverse, log, range, square, square root, and standardize.

Mathematical Functions

Log

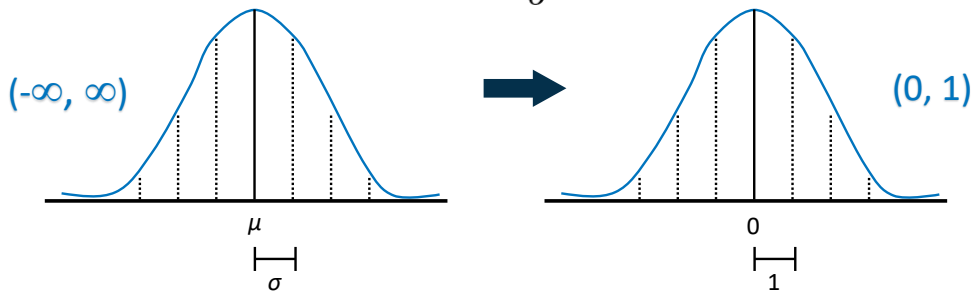


A distribution that is symmetric, or nearly so, is often easier to handle and interpret than a skewed distribution. In the distribution shown here, the variable is skewed to the right, meaning the right-hand tail is stretched out. Applying a log transformation to the input can reduce the skewness, thus making the shape more symmetric.

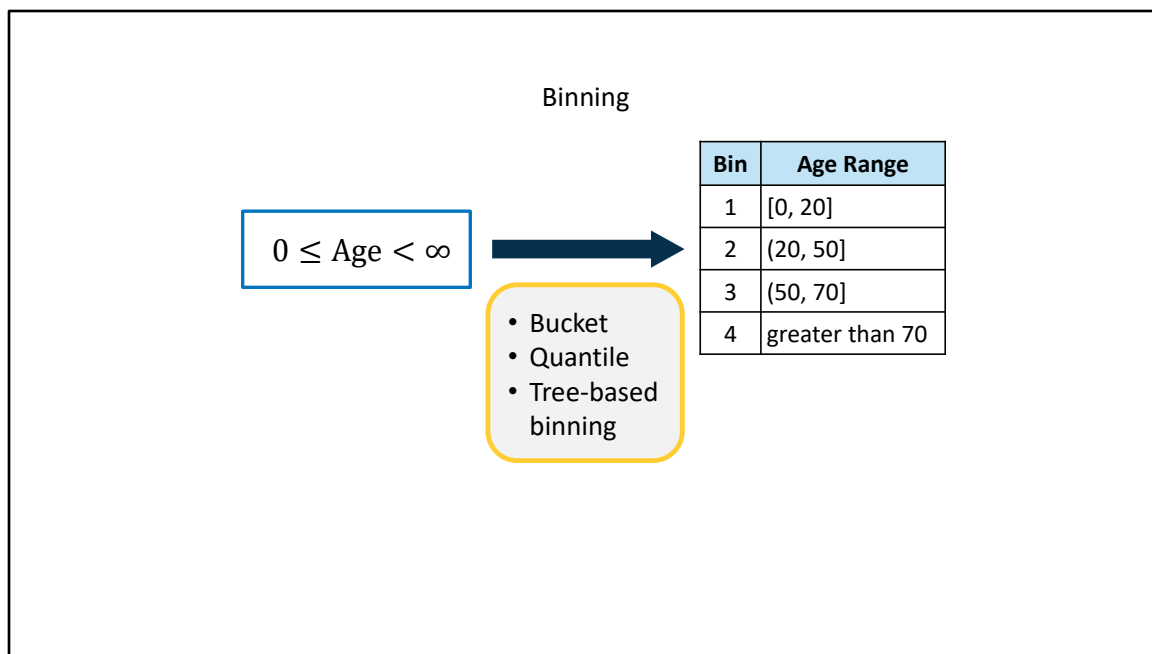
Mathematical Functions

Standardize

$$Z = \frac{x - \mu}{\sigma}$$

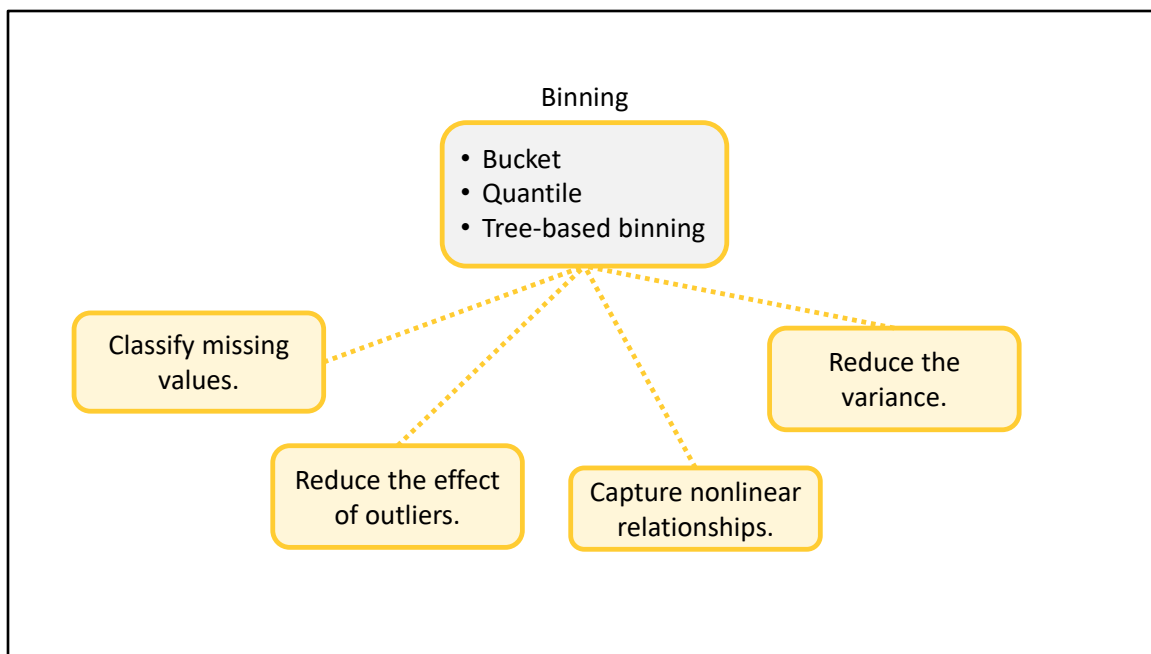


Another method for handling extreme distributions is to use a mathematical function to standardize the inputs. A statistical Z-standardization removes the scale of an input so that it is centered at zero and has a variance of 1. A range standardization transforms an input so that the values range from zero to 1.



Binning is a method of transformation that converts numeric inputs to categories or groups the levels of a high-cardinality input (that is, a categorical input with many levels). In the example shown here, Age is a variable that might theoretically range from zero to infinity. A binning transformation converts Age to an ordinal variable that has only four values, which are represented by bins one through four.

There are many types of binning transformations. In bucketing, the bins themselves are of equal width but the frequency count within each bin varies. Quantile binning creates bins of different widths, but the frequency count of observations is consistent, or at least similar, across the bins. Binning can even be done using a decision tree.



You can use binning to classify missing values, reduce the effect of outliers on a model, or capture nonlinear relationships. A binned version of a variable also has a smaller variance than the original numeric variable.

Transformations in Model Studio

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Question: Transformations



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Question: Transformations

Transformations can do which of the following? (Select all that apply.)

- a. change the shape of the distribution of a variable
- b. reduce the effect of outliers or heavy tails.
- c. standardize inputs to be on the same range or scale.
- d. reduce the bias in model predictions

Answer: Transformations

Transformations can do which of the following? (Select all that apply.)

- ☒ a. change the shape of the distribution of a variable
- ☒ b. reduce the effect of outliers or heavy tails.
- ☒ c. standardize inputs to be on the same range or scale.
- ☒ d. reduce the bias in model predictions

CA: a,b,c,d

FB: Transformations can do all of these things.

Demo: Transforming Inputs

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Question: Transformation Metadata



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Question: Transformation Metadata

When you specify a transformation as metadata for specific variables on the Data tab, that metadata rule overrides the default method specified in the Default interval inputs method property (None) for the Transformations node.

- a. True
- b. False

Answer: Transformation Metadata

When you specify a transformation as metadata for specific variables on the Data tab, that metadata rule overrides the default method specified in the Default interval inputs method property (None) for the Transformations node.

- ☒ a. True
- ☐ b. False

CA: a

FB: This statement is true. It is recommended that you specify transformations for specific variables on the Data tab (or in the Manage Variables node) instead of updating the method in the Transformation node's properties.

2.4 Selecting Features



Selecting Useful Inputs



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Selecting Useful Inputs



Essential Data Tasks

- ✓ Divide the data.
- ✓ Address rare events.
- ✓ Manage missing values.
- ✓ Replace incorrect values.
- ✓ Add unstructured data.
- ✓ Extract features.
- ✓ Handle extreme or unusual values.
- Select useful inputs. feature selection

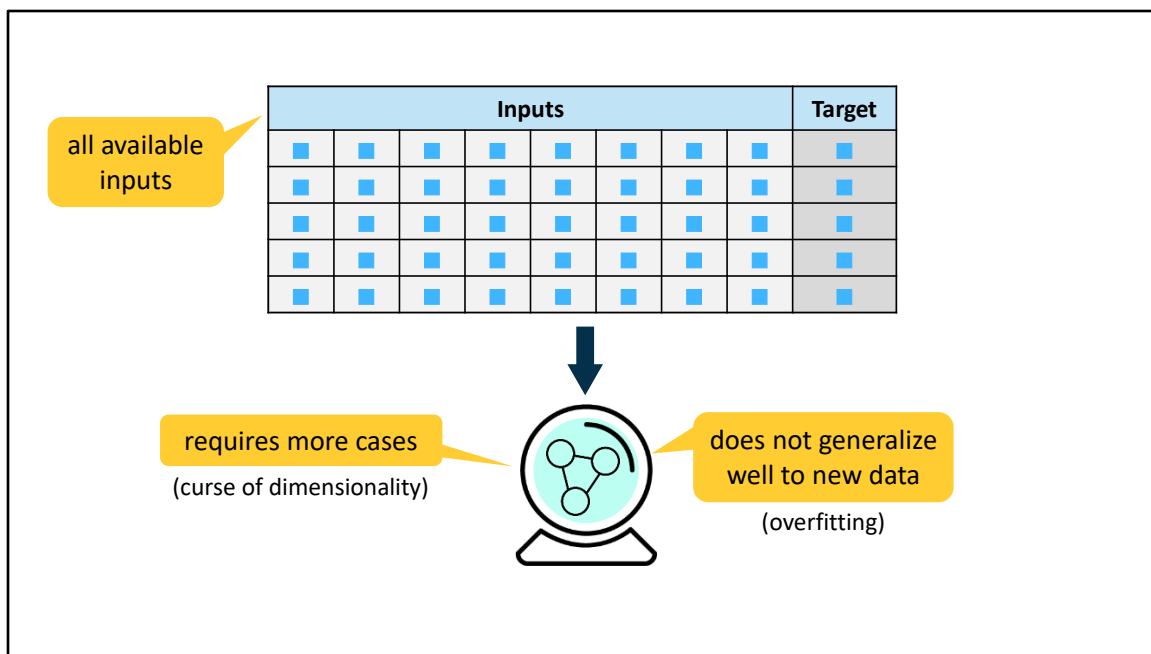
Selecting useful inputs (a process called feature selection) is an essential step in data preprocessing.

Inputs								Target
■	■	■	■	■	■	■	■	■
■	■	■	■	■	■	■	■	■
■	■	■	■	■	■	■	■	■
■	■	■	■	■	■	■	■	■
■	■	■	■	■	■	■	■	■

Use a subset of the input variables in the model.

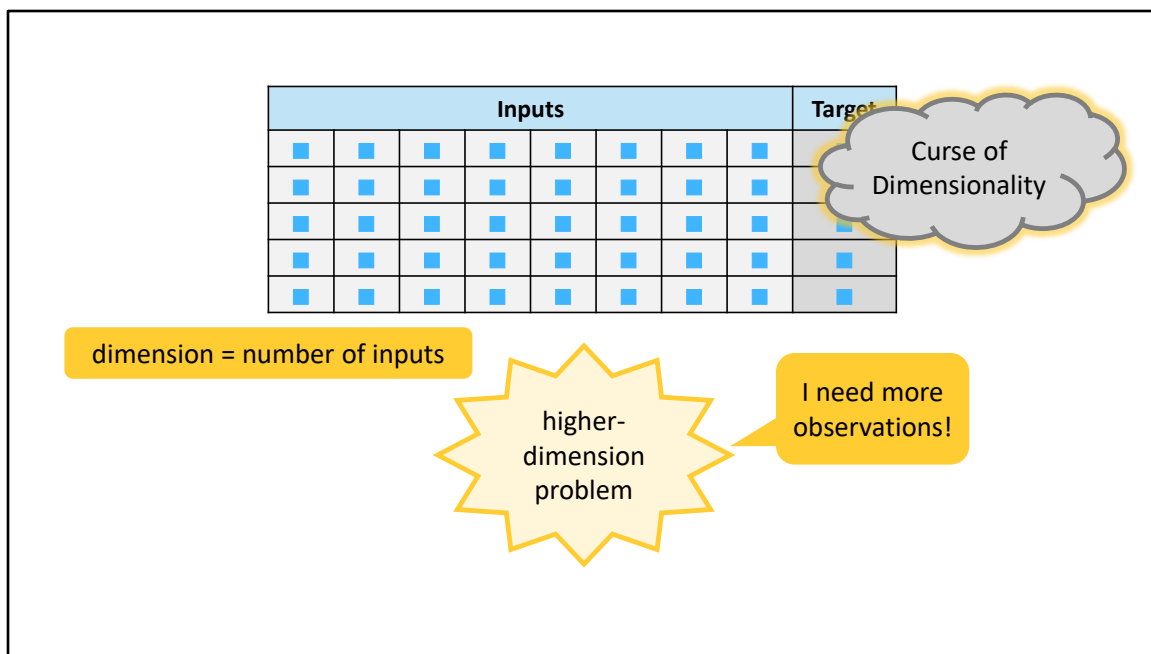


It is important to use a subset of the input variables in the final model for several reasons.



First, remember that using all available inputs almost always results in a predictive model that is overfit to the training data and does not generalize well to new data.

Also, the more inputs you use to build the model, the more cases are required to discover the relationship between the inputs and the target. Let's see why.



The dimension of a data mining problem refers to the number of input variables in the data. The raw data often have a large number of input variables, so data mining problems often have massive dimensions to start.

Higher-dimension problems require a larger number of observations to densely populate the space that represents the relationship between the inputs and the target. This is known as the curse of dimensionality.

Inputs								Target
■	■	■	■	■	■	■	■	■
■	■	■	■	■	■	■	■	■
■	■	■	■	■	■	■	■	■
■	■	■	■	■	■	■	■	■
■	■	■	■	■	■	■	■	■

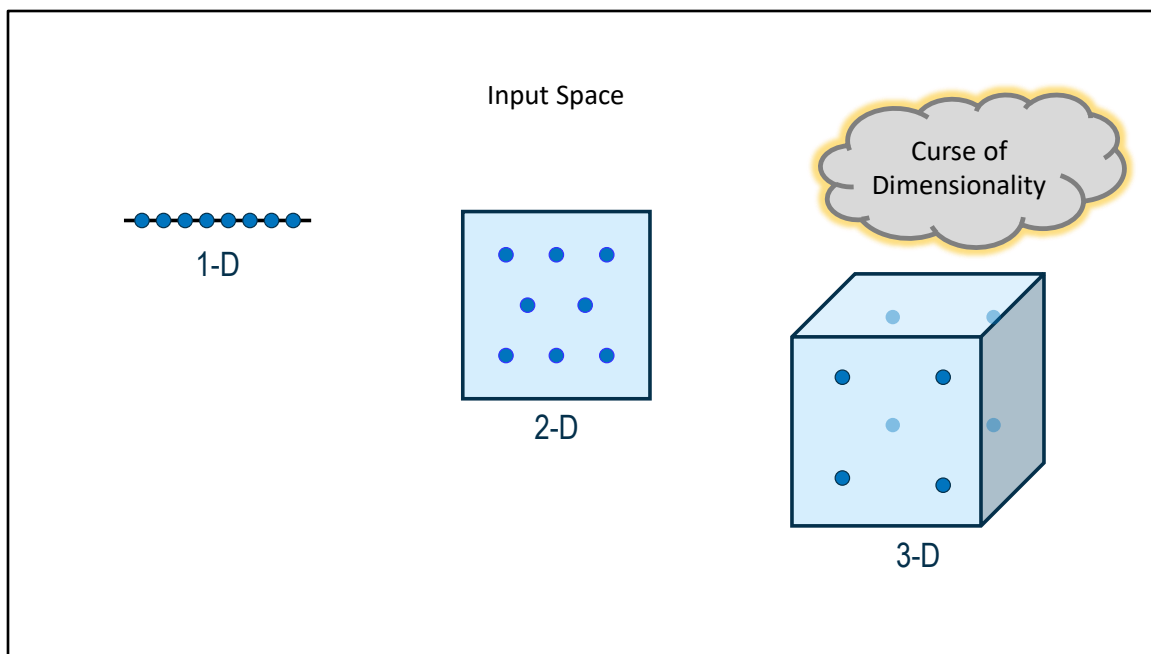
Curse of Dimensionality

highly complex model

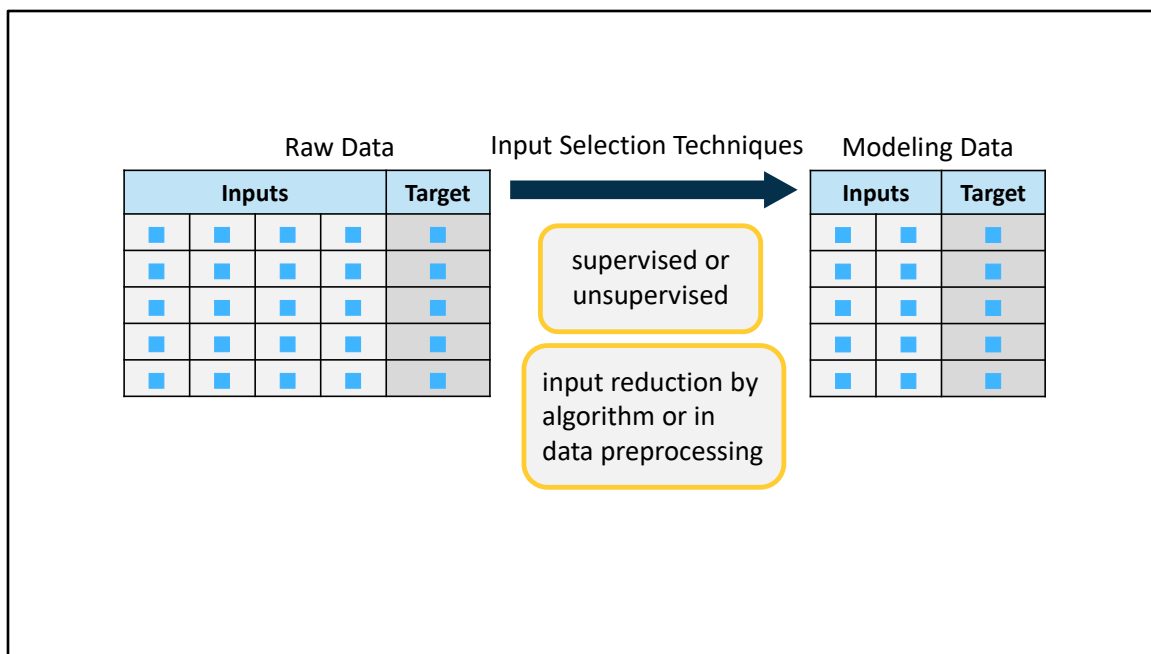


I need a densely populated input space!

It is especially important to have a densely populated input space when fitting highly complex models.

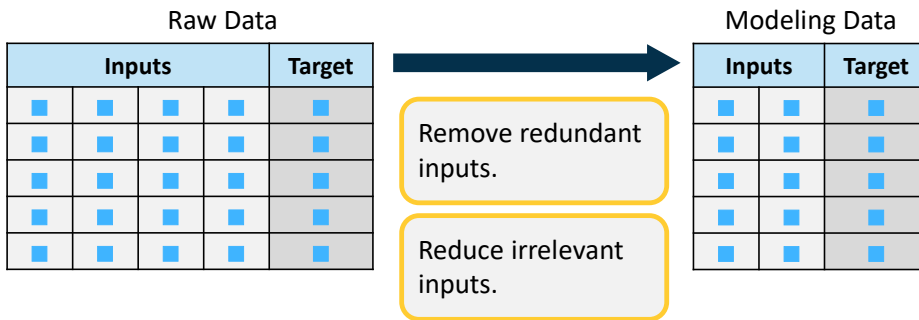


The basic example shown here assumes eight observations. The eight data points very densely fill a one-dimensional input space (that is, a line). These eight points less densely fill a two-dimensional input space (that is, a plane), and even less densely fill a three-dimensional input space (that is, a cube). Imagine how sparse the same eight data points would be in a 10-dimensional, or even 100-dimensional input space. This example shows why it is important to select a subset of inputs.

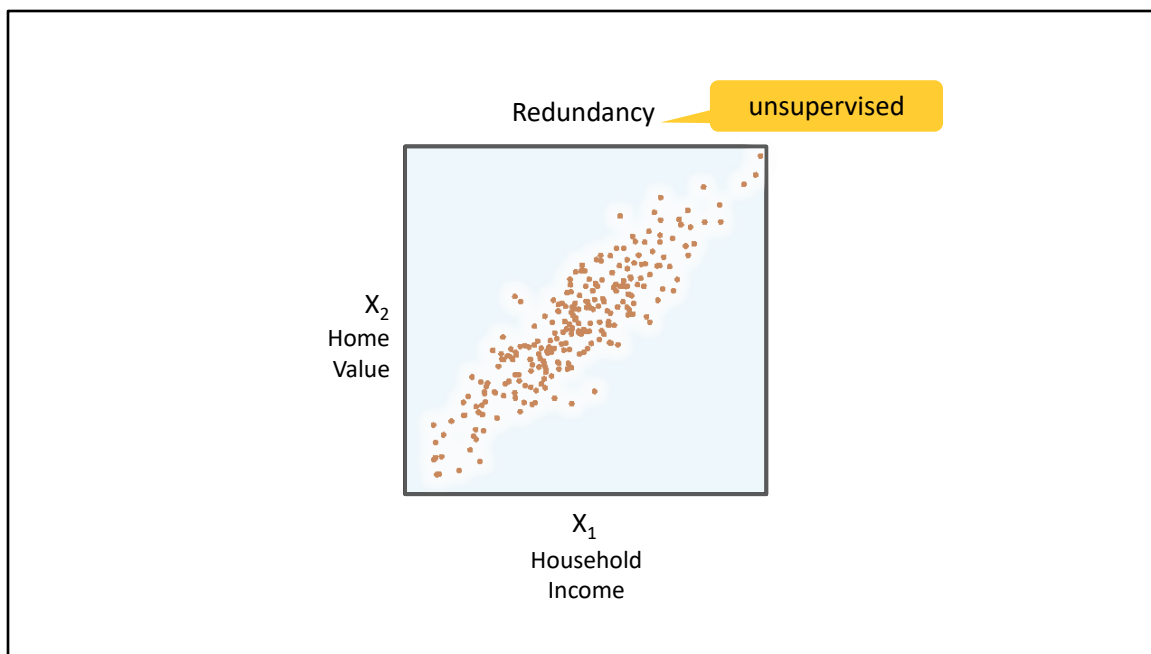


There are many techniques for selecting inputs for a model. Some of these methods are supervised, which means that the target variable is used in the selection process. Other methods are unsupervised and ignore the target. Further, during model building, some modeling algorithms reduce the number of inputs (for example, decision trees), while other algorithms do not (for example, neural networks). For algorithms that do not reduce the number of inputs, it is especially important to reduce the number of inputs during data preprocessing.

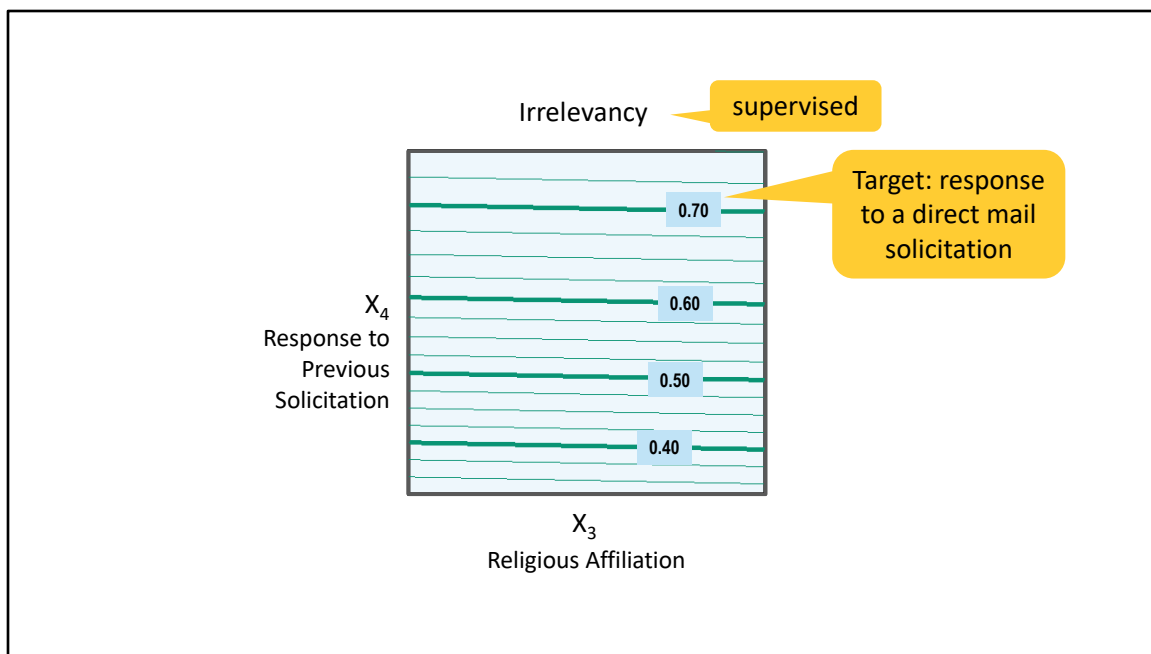
Do not disregard important information!



When we reduce the dimensionality of a problem by selecting inputs, we need to be careful not to disregard important information. Two ways to avoid this are to remove redundant inputs or reduce irrelevant inputs. Let's take a quick look at each method.



A *redundant* input does not give any new information (that is, information that was not already explained by other inputs). In the example shown here, the inputs X_1 and X_2 are highly correlated, so the value of input X_1 gives you a good idea of the value of X_2 . In a business context, two such variables might be household income and home value. Inputs that are highly correlated should not appear in the same model. The target is not used to determine redundant inputs, so redundancy is an unsupervised concept.



An *irrelevant* input does not provide information about the target. In the example shown here, the target values change with changes in X_4 , but not with changes in X_3 .

A business example of irrelevancy is the relationship between religious affiliation and response to a direct mail solicitation. Religious affiliation does not change the probability of response to a direct mail solicitation. However, the response to a previous solicitation does change the probability of responding to a current mail solicitation. If an input is irrelevant to a given target, there is clearly no reason to have it in a model. Irrelevancy relates to whether a variable affects the target, so that it is a supervised concept.

Demo: Selecting Features

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Question: Variable Selection Node



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Question: Variable Selection Node

In the results from the Variable Selection node is a Variable Selection table with a Reason column. What does it mean when the Reason column for an input is blank?

- a. The input was selected.
- b. The input was rejected.
- c. The node did not have enough information to decide whether to select or reject the input.

Answer: Variable Selection Node

In the results from the Variable Selection node is a Variable Selection table with a Reason column. What does it mean when the Reason column for an input is blank?

- a. The input was selected.
- b. The input was rejected.
- c. The node did not have enough information to decide whether to select or reject the input.

CA: a

FB: A blank **Reason** column indicates that the input was selected.

Feature Selection and the Variable Selection Node: Details

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Demo: Saving a Pipeline to the Exchange

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2.5 Best Practices in Data Preparation



Best Practices for Common Data Preparation Challenges

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Question: Data Collection Challenges



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Question: Data Collection Challenges

Biased and incomplete data, sparsity, and high-dimensionality are common data collection challenges. What should you do to overcome these challenges? (Select all that apply.)

Hint: Refer to **Best Practices for Common Data Preparation Challenges**.

- a. Take time to understand the business problem and its context.
- b. Enrich the data.
- c. Select or extract features (or do both) to reduce data dimensions.
- d. Change the representation of the data by applying appropriate transformations.

Answer: Data Collection Challenges

Biased and incomplete data, sparsity, and high-dimensionality are common data collection challenges. What should you do to overcome these challenges? (Select all that apply.)

Hint: Refer to **Best Practices for Common Data Preparation Challenges**.

- ☒ a. Take time to understand the business problem and its context.
- ☒ b. Enrich the data.
- ☒ c. Select or extract features (or do both) to reduce data dimensions.
- ☒ d. Change the representation of the data by applying appropriate transformations.

CA: a,b,c,d

FB: All of the listed actions are recommended for these common data collection challenges.

Feature Engineering in SAS Viya

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2.6 Selecting an Algorithm



Starting the Discovery Phase of the Analytics Life Cycle



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Starting the Discovery Phase of the Analytics Life Cycle



Essential Data Tasks

- ✓ Divide the data.
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- ✓ Extract features.
- ✓ Handle extreme or unusual values.
- ✓ Select useful inputs.

You prepared the data for analysis.



Essential Discovery Tasks

- Select an algorithm.
- Improve the model.
- Optimize the complexity of the model.
- Regularize and tune the hyperparameters of the model.
- Build ensemble models.
- Attempt other algorithms.

Now you're moving into the Discovery phase of the analytics life cycle. There are several essential tasks for discovery. The first one is to select an algorithm. Keep in mind there is no single model that works best for every problem. If that were the case, you would need to use only one algorithm! Take the time to experiment with different algorithms and different modeling options until you find the best model for your particular application.

In this course, you learn how to build several types of machine learning models. For each model, you get to try out different settings in your search for the best model. You also have a chance to compare the models that you built.

The next few resources provide general considerations for selecting an algorithm as well as details about a variety of predictive models. You'll likely want to skim these resources now, and then return to them later, after you've had experience building each type of model. This course can serve as a reference later.

Supervised Learning Algorithms in Model Studio

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Considerations for Selecting an Algorithm

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Question: Algorithm Selection



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Discussion

Suppose you are assigned to build a model for the marketing team in your company. They want to use a data-driven approach to design an effective direct marketing campaign. They also want to understand the relationship between various marketing attributes and the likelihood that a customer will purchase a product. You will work with a large data set that contains information for all customers who have bought a product since the company was started five years ago. You have not seen the data yet. The project deadline is the end of the second quarter, which is less than two months away.

What are some key considerations for selecting algorithms? Which algorithms might be appropriate for this project, based on what you know?



FB: When selecting algorithms for this project, you might want to consider the following:

Size and nature of the data: You know that the data set is large. Linear models can be a good choice for large data sets because of their training efficiency and the curse of dimensionality. However, you have not had a chance to explore the data, so you do not know whether the relationship between the inputs and the target is linear. Therefore, you cannot rule out the more sophisticated machine learning algorithms.

Analysis goal: The marketing team wants to design an effective direct marketing campaign, which involves predicting whether customers will make a purchase (the target). If prediction were the only goal, and barring any time constraints, you would want to consider the most sophisticated algorithm that you could afford in terms of time and computational resources. However, remember that the project also has a second goal: to understand the relationship between the inputs (various marketing attributes) and the target. This second goal requires a model that is interpretable, which narrows your choices to regression or decision trees.

Time to build the model: The deadline for the project is less than two months away. After you explore and prepare the data, you will have a better idea of how much time is left to build models.

These are only a few of the considerations for selecting algorithms. As you learn more about a project, other considerations might arise.

Comparison of Modeling Algorithms

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2.7 Lesson Conclusion



Lesson Quiz

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